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Project Proposal

The CDC Diabetes Health Indicators dataset represents healthcare statistics and lifestyle information about individuals and their diabetes diagnoses. According to the archive UC Irvine website, this data set contains 35 features that consist of demographics, lab results, survey questions, and answers from each patient, with each row corresponding to a separate individual in this study. In this project I plan to use the following variables; BMI (Body Mass Index; a measurement of body fat percentage based off of height and weight), Age (Age category, with 13 levels and the numbers 1-13 representing different age groups), PhysActivity (Physical activity measured on the previous 30 days, not including work.), HighBP (blood pressure rated on a scale of 0 being not high, and 1 being high), HighChol (High cholesterol with 0 being not high and 1 being high), HeartDiseaseorAttack (history of heart disease indicated CHD or myocardial infarction with 0 being no and 1 being yes), and education (Education level based on a scale of 1-6, 1 being never attended school or only kindergarten, 2 being grades 1-8(elementary), 3 being grades 9-11(Highschool), 4 being grade 12 or GED(senior or alternate graduation), 5 being college 1-3 years, 6 being 4 years or more of college).

There are endless questions this dataset can answer, but one that piques my interest is: in what ways does a history of heart risk factors, such as high blood pressure, high cholesterol, and heart disease, affect the likelihood of an individual being diagnosed with diabetes? This research

question is relevant because, according to the CDC, approximately 40.1 million Americans, or almost twelve percent of the U.S. population, currently have diabetes, and it is a major public health issue. (National Diabetes Statistic report 1) This research can identify the most significant risk factors for individuals and inform what people, as well as doctors, should work on to lower the risk of developing diabetes and potentially prevent it.

Five hypotheses I am considering to further explore in depth, based on my research question, are the following:

1. There is a significant difference in the BMI of individuals who have diabetes compared to those without.
2. There is a significant difference in those with diabetes compared to those without when it comes to having heart issues and those with none.
3. There is a positive, statistically significant relationship between a history of heart disease and high blood pressure.
4. There is a negative, statistically significant relationship between physical activity and the possibility of being diagnosed with diabetes.
5. There is a positive, statistically significant relationship between BMI and the likelihood of having a history of heart disease or a heart attack.

Works Cited

Centers for Disease Control and Prevention. "Data & Statistics." *CDC*, 13 June 2026,

www.cdc.gov/diabetes/php/data-research/index.html.

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Irvine, 25 Sept. 2023, archive.ics.uci.edu/dataset/891/cdc+diabetes+health+indicators